SIMPLE CRUD API (WITHOUT DATABASE) – STUDENTS DATA

```
const express = require("express");
const app = express();
const PORT = 3000;
// Middleware to read JSON data
app.use(express.json());

let students = [
   { id: 1, name: "Priya", course: "BCA" },
   { id: 2, name: "Rahul", course: "BCA" }
];

// GET - Get all students
app.get("/students", (req, res) => {
   res.json(students);
});

// GET - Get one student
app.get("/students/:id", (req, res) => {
   const id = parseInt(req.params.id);
   const student = students.find(s => s.id === id);
   if (!student) {
      return res.status(404).json({
         message: "Student not found"
      });
   }
   res.json(student);
});

// POST - Add a new student
app.post("/students", (req, res) => {
   const newStudent = {
      id: students.length + 1,
```

```
        name: req.body.name,
        course: req.body.course
    };
    students.push(newStudent);
    res.status(201).json({
        message: "Student added successfully",
        student: newStudent
    });
});

// PUT - Update a student
app.put("/students/:id", (req, res) => {
    const id = parseInt(req.params.id);
    const student = students.find(s => s.id === id);
    if (!student) {
        return res.status(404).json({
            message: "Student not found"
        });
    }
    student.name = req.body.name;
    student.course = req.body.course;
    res.json({
        message: "Student updated successfully",
        student: student
    });
});

// DELETE - Delete a student
app.delete("/students/:id", (req, res) => {
    const id = parseInt(req.params.id);
    const index = students.findIndex(s => s.id === id);
    if (index === -1) {
        return res.status(404).json({
```

```
            message: "Student not found"
        });
    }
    students.splice(index, 1);
    res.json({
        message: "Student deleted successfully"
    });
});

// Start server
app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```

CRUD API (DATABASE) – STUDENTS DATA

```
const express = require("express");
const mysql = require("mysql2");
const app = express();
const PORT = 3000;
// Middleware
app.use(express.json());

const db = mysql.createConnection({
    host: "localhost",
    user: "root",
    password: "",
    database: "studentdb"});
db.connect((err) => {
    if (err) {
        console.log("Database connection failed:", err.message);
        return;}
    console.log("MySQL connected successfully");
```

```
});

// GET - All students
app.get("/students", (req, res) => {
    const sql = "SELECT * FROM students";
    db.query(sql, (err, results) => {
        if (err) {
            return res.status(500).json({
                error: err.message
            });
        }
        res.json(results);
    });
});

// GET - One student
app.get("/students/:id", (req, res) => {
    const id = req.params.id;
    const sql = "SELECT * FROM students WHERE id = ?";
    db.query(sql, [id], (err, results) => {
        if (err) {
            return res.status(500).json({
                error: err.message
            });
        }
        if (results.length === 0) {
            return res.status(404).json({
                message: "Student not found"
            });
        }
        res.json(results[0]);
    });
});
```

```
// POST - Add student
app.post("/students", (req, res) => {
    const { name, email, course } = req.body;
    const sql = `
        INSERT INTO students (name, email, course)
        VALUES (?, ?, ?)
    `;
    db.query(sql, [name, email, course], (err, result) => {
        if (err) {
            return res.status(500).json({
                error: err.message
            });
        }
        res.status(201).json({
            message: "Student added successfully",
            id: result.insertId
        });
    });
});

// PUT - Update student
app.put("/students/:id", (req, res) => {
    const id = req.params.id;
    const { name, email, course } = req.body;
    const sql = `
        UPDATE students
        SET name = ?, email = ?, course = ?
        WHERE id = ?
    `;
       db.query(sql,[name, email, course, id],(err, result) => {
            if (err) {
                return res.status(500).json({
```

```
                error: err.message
            });
        }
        if (result.affectedRows === 0) {
            return res.status(404).json({
                message: "Student not found"
            });
        }
        res.json({
            message: "Student updated successfully"
        });
    }
  );
});

// DELETE - Delete student
app.delete("/students/:id", (req, res) => {
  const id = req.params.id;
  const sql = "DELETE FROM students WHERE id = ?";
  db.query(sql, [id], (err, result) => {
    if (err) {
        return res.status(500).json({
            error: err.message
        });
    }
    if (result.affectedRows === 0) {
        return res.status(404).json({
            message: "Student not found"
        });
    }
    res.json({
        message: "Student deleted successfully"
    });
```

```
    });
});
// Start server
app.listen(PORT, () => {
    console.log(`Server running at http://localhost:${PORT}`);
});
```